

The Fenchel-Neilsen Construction of Compact Hyperbolic Surfaces

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So far, we've seen that we can construct a large number of hyperbolic surfaces by quotienting $\mathbb{H}^2$ by a sufficiently nice discrete group Γ ; in fact, we saw that all complete hyperbolic surfaces can be obtained this way. The purpose of this talk is give a detailed exposition of a different, ground-up approach to constructing hyperbolic surfaces. We'll also touch a bit on the uniqueness of our constructions.

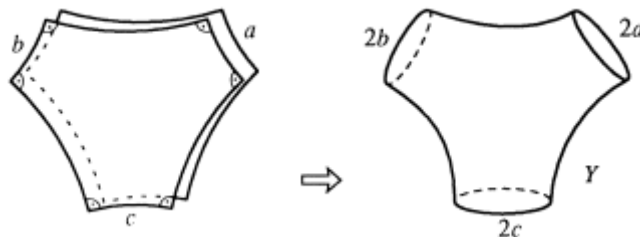
I. The Fenchel-Neilsen Parameters

Fix a non-negative integer g and suppose we want to construct a hyperbolic surface of genus g . In fact, let's suppose that we also would like to be able to "reference" this surface uniquely through a nice parameter space that's easy to visualize. That's the purpose of this section; namely, we outline a method of constructing hyperbolic surfaces developed by Fenchel and Nielsen in the mid-1900s, the main idea behind which is that we can build (and bookkeep) a variety of hyperbolic surfaces by gluing together many copies of one basic unit: the beloved *pair of pants* or *Y-piece*:

Definition. A *Y-piece* (or *pair of pants*) is a hyperbolic surface of signature $(0, 3)$.

Remark. Recall that a compact connected orientable topological surface with boundary is said to have signature (g, n) if it is obtained from a closed surface of genus g by cutting out the interior n disjoint closed disks.

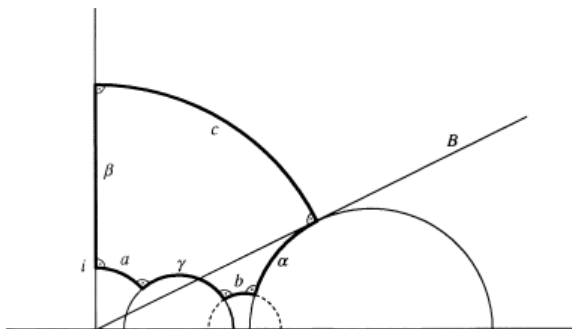
The first step in the construction is to actually build our building blocks. In particular, given any three positive numbers a , b , and c , can we find a *Y-piece* with these numbers as the lengths of its boundary geodesics? It may seem obvious, but let's prove it. Note that we can obtain a *Y-piece* by gluing together two copies of a hyperbolic right-angled hexagon H along its edges, as depicted in the figure below:



Because each boundary geodesic shall have length twice that of its corresponding edge, it shall suffice to show that given any three positive numbers a , b , and c , we can find a right-angled hexagon with pairwise non-adjacent sides of these lengths. We do so below.

Lemma 1.1. *For any three positive numbers a , b , and c , there exists a right-angled hexagon in the hyperbolic plane with pairwise non-adjacent sides of length a , b , and c .*

Proof. Our model is the upper half-plane $\mathbb{H}^2$. Consider a vertical geodesic of length β situated along the y -axis, and consider a geodesic of length γ parallel to the positive x -axis. Translate γ along the x -axis until the common perpendicular between β and γ has length a . Now let B be the line consisting of all points in $\mathbb{H}^2$ of distance c from β which lie on the same side of β as γ (note that B shall be a Euclidean straight line because the mappings $z \mapsto \lambda z$ for $\lambda > 0$ are hyperbolic isometries). Finally, consider a moving geodesic α tangent to B . Move α along B until $d(\gamma, \alpha) = b$. Drawing the common perpendicular from γ to α then completes the hexagon. See the figure below.

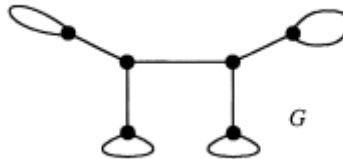


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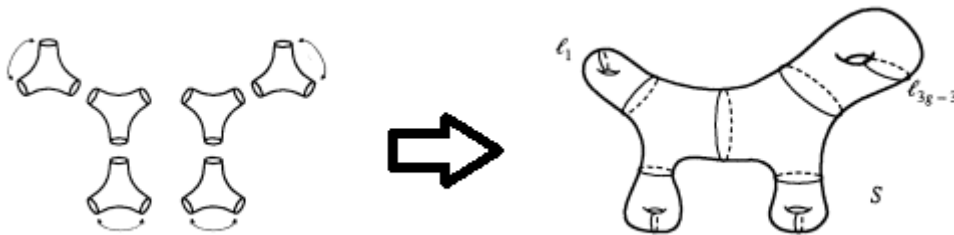
Corollary 1.1. *For any three positive numbers a , b , and c , there exists a Y -piece whose boundary geodesics are of length a , b , and c .*

Remark. In fact, this is the *unique* such Y -piece. For let Y and $\tilde{Y}$ be two such pieces. For each pair of boundary geodesics on Y (resp. $\tilde{Y}$), there is a unique common perpendicular; together, these perpendiculars decompose Y (resp. $\tilde{Y}$) into two isometric right-angled hexagons. In fact, it is not hard to see that the endpoints of these perpendiculars divide each boundary geodesic of Y (resp. $\tilde{Y}$) into two arcs of the same length, so all four of the hexagons we get from decomposing Y and $\tilde{Y}$ are isometric; together, they give an isometry $Y \rightarrow \tilde{Y}$.

Now consider a cubic graph G (by this I just mean a connected graph that is 3-regular) with $2g - 2$ vertices and $3g - 3$ edges; for instance, if $g = 4$ then G may look something like



You should think of G as a sort of blueprint for our surface; in particular, we associate to each vertex in G a Y -piece, and paste the pieces together if there is an edge connecting their associated vertices. In fact, thought about in this way, the numbers $2g - 2$ and $3g - 3$ of vertices and edges (which might have seemed an odd choice at first) are not so mystical; they come from the Euler characteristic a non-punctured surface of genus g . In the example above, we get



In other words, we've successfully constructed a genus g surface which inherits a hyperbolic structure from its building blocks. Note that boundary geodesics that come together must have the same length, so on the resulting surface S we obtain $3g - 3$ closed geodesics (one for each edge in G). By Lemma 1.1, the lengths $\ell_1, \dots, \ell_{3g-3}$ of these geodesics can be anything. However, there's also one more thing that we can do: we can prescribe "twist parameters" $\alpha_1, \dots, \alpha_{3g-3}$ for the pasting of each of the closed geodesics. These are real numbers that provide an offset to the pasting condition so that geometrically we get a twist in our surface. For instance, say that we wanted to paste together the boundary geodesics γ_1 and γ_2 , each parameterized on the unit interval $[0, 1]$. We can lift these parameterizations to $\mathbb{R}$ by declaring $\gamma_1(t) = \gamma_1(t - n)$ for $t \in [n, n + 1]$ and similarly for γ_2 . The specification of a twist parameter α then says that we should paste γ_1 and γ_2 together according to

$$\gamma_1(t) \sim \gamma_2(\alpha - t), \quad t \in \mathbb{R}.$$

Together, the parameters $\ell_1, \dots, \ell_{3g-3}, \alpha_1, \dots, \alpha_{3g-3}$ determine our surface S uniquely (in fact, this uniqueness immediate from the construction), and are called the *Fenchel-Nielsen* parameters of S . Note that they give rise to a $(6g - 6)$ -dimensional space of examples of (compact) hyperbolic surfaces of genus g . In fact, we'll later see that *all* compact hyperbolic surfaces without boundary of genus at least two may be constructed in this fashion.

II. Marked Y -Pieces

From Corollary 1.1, one might be tempted to conclude that $\mathbb{R}_+^3$ serves as a model for the space of all Y -pieces. This *is* true insofar as each point in $\mathbb{R}_+^3$ specifies a Y -piece, however it is not an adequate model for the space

of all such pieces because this correspondence is not one-to-one. The reason for this is that $\mathbb{R}_+^3$ distinguishes between the points $(1, 2, 1)$, $(1, 1, 2)$, and $(2, 1, 1)$, for instance, despite the fact that all these points are associated to the same Y -piece. An easy remedy to this is to introduce *markings* to our Y -pieces. A marking for a Y -piece Y is simply a choice of labelling for each boundary geodesic by the integers 1, 2, and 3. In this case, we say that two Y -pieces Y and $\tilde{Y}$ with labelled boundary geodesics $\gamma_1, \gamma_2, \gamma_3$ and $\tilde{\gamma}_1, \tilde{\gamma}_2, \tilde{\gamma}_3$ are *marking equivalent* if there is an isometry $\varphi: Y \rightarrow \tilde{Y}$ which sends γ_i to $\tilde{\gamma}_i$ for each i . Notice that $(1, 2, 1)$, $(1, 1, 2)$, and $(2, 1, 1)$ all determine different *marked* Y -pieces, so the correspondence between $\mathbb{R}_+^3$ and *marked* Y -pieces is in fact one-to-one. One should think markings as a way of keeping track of extra data.

III. Constructing General Compact Hyperbolic Surfaces

We saw in the first section that we can build construct a good amount of compact hyperbolic surfaces via the Fenchel-Neilsen method. It turns out, however, that *all* compact hyperbolic surfaces without boundary of genus at least two can be constructed this way. We now detail a proof of this.

Lemma 2.1. *Let $\varphi: S \rightarrow R$ be a homeomorphism of genus g compact hyperbolic surfaces without boundary, and let $\gamma_1, \dots, \gamma_n$ be pairwise disjoint, simple closed geodesics on S . Then there is a homeomorphism $\Phi: S \rightarrow R$ isotopic to φ such that the curves $\Phi(\gamma_1), \dots, \Phi(\gamma_n)$ are each simple closed geodesics on R .*

Theorem 2.1. *Let S be a compact hyperbolic surface without boundary of genus $g \geq 2$. Then there exist $\ell_1, \dots, \ell_{3g-3} \in \mathbb{R}_+$ and $\alpha_1, \dots, \alpha_{3g-3} \in R$ such that S is isometric to the unique hyperbolic surface with Fenchel-Neilsen parameters $(\ell_1, \dots, \ell_{3g-3}, \alpha_1, \dots, \alpha_{3g-3})$.*

Proof. From the first section, we know that we can construct a compact hyperbolic surface of genus g via the Fenchel-Neilsen method; fix any such space S_0 . Now let's forget about the hyperbolic structure on S_0 . From the classification of compact genus g surfaces without boundary, we infer the existence of a homeomorphism $\varphi: S_0 \rightarrow S$. By the Lemma above, φ can be chosen so that parameter geodesics $\gamma_1, \dots, \gamma_{3g-3}$ of S_0 form a system of simple closed geodesics on S . It follows that cutting S along these geodesics gives a decomposition of S into Y -pieces. These determine S as a surface constructed via the Fenchel-Neilsen process for suitable parameters $\ell_1, \dots, \ell_{3g-3}$ and $\alpha_1, \dots, \alpha_{3g-3}$. $\square$

Remark. Note that the parameters above need not be the same as those of our base surface S_0 , for the homeomorphism φ knows nothing about the underlying hyperbolic structures on S and S_0 . What φ *does* tell us, however, is that as hyperbolic surfaces, S_0 and S can be constructed from the same *blueprint* cubic graph G .